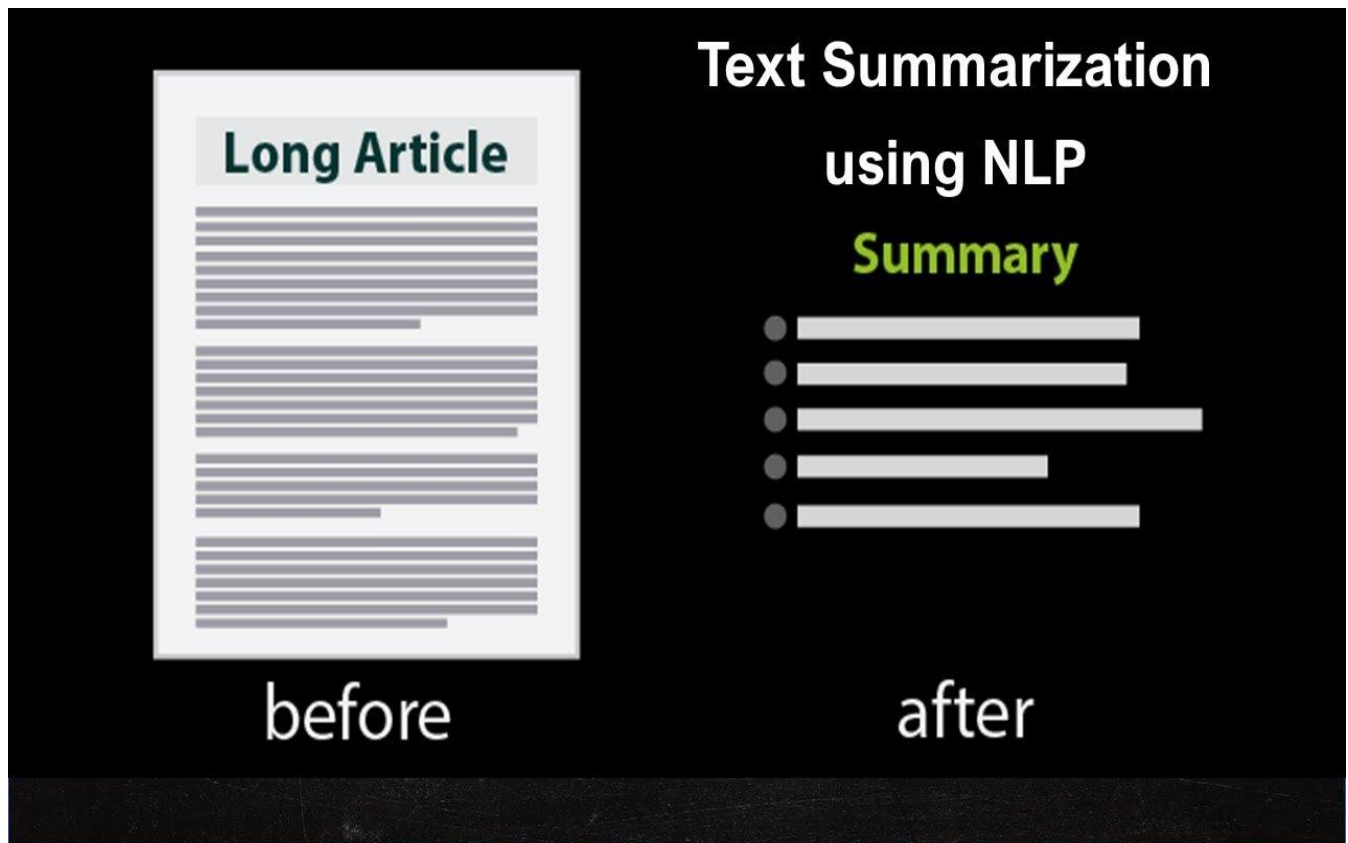




Project name: Text Summarize

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1. Project Idea Summary

The project focuses on summarizing lengthy news articles using extractive summarization. The approach identifies the most important sentences in an article by analyzing how similar they are to other sentences within the same article. The goal is to produce summaries that capture the article's key points without generating new text. This method is practical for summarizing large text datasets efficiently. This method is effective for summarization tasks that don't require deep contextual understanding or language generation, making it computationally efficient and scalable.

2. Dataset Used

- **Name:** abisee/cnn_dailymail
 - **Description:** This dataset contains news articles and their summaries, primarily used for text summarization tasks. Each entry includes an article and a human-written summary, making it suitable for both extractive and abstractive summarization models.
 - **Source Website:** [Hugging Face Datasets - abisee/cnn_dailymail](#)
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3. Model Summary

The project uses **TF-IDF Vectorization** and **Cosine Similarity**, following an **unsupervised extractive summarization** approach:

Model Steps:

1. **Text Preprocessing:**
 - Splits the article into sentences using punctuation as delimiters.
2. **TF-IDF Vectorization:**
 - Converts each sentence into a vector of word importance scores.
 - **TF (Term Frequency):** Measures how often a word appears in a sentence.
 - **IDF (Inverse Document Frequency):** Reduces the importance of common words by assigning lower scores to frequently occurring terms like "the" or "is."

3. **Cosine Similarity Calculation:**

- Computes similarity between all pairs of sentences.
- Higher similarity indicates that sentences share more important content.

4. **Sentence Ranking:**

- Aggregates similarity scores for each sentence.
- Sentences with the highest average similarity are considered central to the article's main points.

5. **Summary Generation:**

- Selects the top 3 highest-scoring sentences to form the summary.
- These sentences are returned in order of relevance.

6. **Data Handling:**

- **Dataset Concatenation:** Combines the training and testing datasets from the Hugging Face dataset to create a unified dataset for processing.
- **Train/Test Split:** Splits the dataset into 80% training data and 20% test data using (`train_test_split`) for model evaluation.

7. **Output Generation:**

- **DataFrame Creation:** Creates a DataFrame containing the original articles and their corresponding summaries.
- **CSV Saving:** Saves the summaries to a CSV file, separating them into training and test sets for further analysis.

8. **Final Summary:**

- After processing all articles, the summaries for the training and test data are saved in CSV format, making the data easy to inspect or use for model evaluation.
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